Investigating the Latest Status of Spanglish in Multilingual Large Language Models (MLLMs)

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Abstract
Multilingual Large Language Models (MLLMs) have made significant advancements in natural language processing (NLP), demonstrating improvement in understanding code-switched (CSW) text and developing novel datasets to bridge the gaps found in low-resource languages. However, these models continue to struggle with CSW languages, such as Spanglish, which is widely spoken in English-Spanish communities. Although models like PANGEA-7B demonstrate improved bilingual capabilities through their fine-tuned English-Spanish training strategies, mixed-language data in the real world remain underrepresented in model development. We first aim to replicate results from PANGEA-7B and then investigate its ability to capture nuanced Spanglish dynamics by evaluating model outputs. Then, we provide an overview of the challenges, limitations, and workarounds presented by the experiment. The purpose of this investigation is to identify what types of training data and evaluation metrics best support the Spanglish community. In doing so, our objective is to help shape future model design and dataset curation practices to better reflect the complexity and legitimacy of multilingual, multi-dialectal communication.

1 Introduction

As multilingual large language models (MLLMs) evolve, users around the world gain the ability to access high-end models using their native language. However, despite the usage of multilingual datasets during model training, models continue to fail to capture the complexity of code-switching (CSW). CSW is when a speaker switches between two languages within a single sentence or phrase, and the lack of adequate datasets containing real human-annotated CSW conversations leads to an inability in models to comprehensively capture the natural flow of CSW speech. Commonly

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found in Spanglish and other hybrid languages, this phenomenon continues to present challenges in natural language processing (NLP).

Despite the multilingual advances made in NLP, Spanglish remains underrepresented in its selection of human-annotated datasets and evaluation benchmarks. Recent surveys[6] have emphasized that the expansion of multilingualism does not equate to a model's understanding of CSW text, as models often lag behind more specialized or fine-tuned systems[11]. Models may know English and Spanish as monolingual languages, but they fail to model the structural and cultural nuances of how they co-occur in real-world scenarios. They do, however, recognize CSW text on a token level[5].

To address these gaps, new MLLMs have been developed. Using widely accepted dataset curation methods to satisfy the lack of sufficient CSW datasets, PANGEA-7B[10] improves multilingual modeling by fine-tuning English and Spanish training strategies to produce more detailed outputs in tasks such as visual question answering (VQA) and question answering (QA), while SwitchLingua[9] introduces a large-scale multiethnic dataset explicitly focusing on CSW in 12 languages.

This investigation aims to explore the abilities of PANGEA-7B to identify and handle CSW cases in Spanglish. We begin by replicating some of the results from PANGEA-7B's paper and verify the alignment of the output with the original studies. Then, we record our observations and form them into a comprehensive overview of the current limitations that MLLMs are undergoing in the CSW field. Our goal is to identify data types and evaluation metrics that best support Spanglish with the aim of shaping future model design to accurately reflect the characteristics found within the Spanglish hybrid language.

2 Related Work and Background

MLLMs have become increasingly effective in performing particular tasks in dozens of languages, but their performance remains inconsistent when faced with CSW text, especially if the CSW text is not between two languages using a Latin alphabet. CSW, especially in the form of Spanglish, involves mixing English and Spanish in a way that reflects everyday speech between bilingual speakers. Traditional models often overlook this behavior due to the lack of training data, preventing an accurate representation of the linguistic patterns found in daily life.

In order to address this gap, recent studies[11][5] have come out to explain this phenomenon. In these studies, it was argued that most MLLMs are not yet effective code-switchers because of their poor performance on basic CSW benchmarks despite their multilingual training[11]. Similarly, another study showed that while pre-trained models (PLMs) can generalize code-switched inputs when probed carefully, they still lack accurate representations of real-world CSW contexts[5].

SwitchLingua provides the first large-scale, multilingual, and multiethnic dataset that specifically targets CSW. It spans 12 languages and 420K parallel machine-translated data pairs from LinCE[1], offering a rich training ground for models to learn how to accurately represent real-world CSW. It also introduces SAER, a semantic-aware evaluation metric that better captures meaning preservation in CSW text, which is important for hybrid languages such as Spanglish where grammatical alignment is not always possible.

In parallel, PANGEA-7B represents a developed focus on making MLLMs better at performing particular tasks such as VQA and QA. While SwitchLingua focuses more on audio-only tasks and PANGEA-7B focuses on extending tasks to be capable on a multimodal scale, both aim to achieve inclusivity when it comes to multilingual reasoning and translation.

Complementing these approaches, comprehensive surveys[6] exploring the full capabilities of current MLLMs have been published, highlighting the potential and structural limits of current models. Additionally, they explore synthetic data generation for cross-lingual tasks, a strategy that could be adapted to create high-quality Spanglish data in the absence of large, naturally-occurring CSW instances[8].

These works reveal the capabilities and limitations of current MLLMs. While multilingualism has improved dramatically, CSW variants like Spanglish remain underrepresented, falling behind their monolingual successors. This investigation seeks to explore these gaps by evaluating the extent to which tools like PANGEA-7B can model Spanglish and how new types of data or metrics may help align future models more closely with real-world multilingual use.

3 Data and Methodologies

In this study, we first evaluated PANGEA-7B on CSW tasks, specifically focusing on Spanglish. After identifying areas where the model underperformed, we replicated selected results from the original paper to verify our experimental setup. We then expanded our analysis to include massive multitask language understanding (MMLU) tasks while maintaining a focus on CSW evaluation. We conducted our experiments using standardized protocols to ensure consistency and reproducibility across tasks. Performance was assessed using metrics such as accuracy and exact match, with pre-processing applied to datasets to ensure uniform formatting. In order to conduct these experiments, we collected and prepared multiple benchmark datasets, including TyDiQA[7], CohereLabs' Global-MMLU (English and Spanish) dataset[3], and Drew Curran's Spanglish dataset[4].

3.1 Data Collection

To assess PANGEA-7B's performance on both MMLU and CSW tasks, we utilized several publicly available datasets obtained from *Hugging Face*. These resources supported replication of established baseline results as well as evaluation in Spanglish contexts.

1. **TyDiQA**: A multilingual question-answering (QA) dataset covering 11 typologically diverse languages. In this study, it was used to assess reasoning performance and to validate the reliability of our evaluation pipeline.
2. **Global-MMLU**: A benchmark designed to evaluate a model's general reasoning and knowledge. We used the English and Spanish subsets to measure bilingual QA performance in alignment with prior PANGEA-7B MMLU evaluations.
3. **Spanglish dataset**: A machine-translated parallel dataset containing naturally occurring CSW text between English and Spanish, created by Drew Curran. This dataset was used to test PANGEA-7B's capacity to handle CSW scenarios.

All datasets were pre-processed to ensure consistent formatting and tokenization. We used the *Hugging Face* `AutoTokenizer` for tokenization and standardized each dataset's structure to be compatible with our evaluation framework. Dataset versions and access dates were recorded to ensure reproducibility. Any filtering applied, such as removal of incomplete entries or text normalization, was applied consistently across datasets.

3.2 Methods

Our methodology consisted of two primary phases: (1) replication of PANGEA-7B's reported results to validate our experimental setup, and (2) evaluation of its performance on CSW tasks, specifically Spanglish. Across both phases, we followed standardized evaluation protocols to ensure reliability and reproducibility.

3.2.1 Model Setup

The model we used to conduct our studies was PANGEA-7B, an open-source multilingual, multimodal large language model designed to span 39 languages. The model was accessed through *Hugging Face*, and baseline performance was compared with a smaller run using Mistral NeMo[2] to ensure experimental validity and rule out code-related errors.

3.2.2 Evaluation Framework

For the MMLU task, performance was measured using accuracy, while exact match was used for the CSW task. We compared the CSW task outputs against its original reference translations, with alignment assessed at the lexical and syntactic levels.

3.2.3 Prompting Strategies

We experimented with multiple prompting styles to optimize model responses. Initial tests used direct QA prompts, followed by more detailed instructions inspired by chain-of-thought (CoT) prompting. This comparison allowed us to measure sensitivity to prompt structure. PANGEA-7B, as a pattern-based and not an instruction-tuned model, was found to be highly dependent on explicit and rigidly structured prompts.

3.2.4 Experimental Design

- **Replication Phase:** To validate our experimental setup, we first reproduced PANGEA-7B's reported performance on MMLU tasks using the English and Spanish subsets of CohereLabs' Global-MMLU dataset. Initial evaluations of 10 items ensured code correctness, after which the experiments were scaled to full test sets of 100 items.
- **Code-Switching Phase:** Using the Spanglish dataset, we tasked the model with translating English sentences into Spanglish while maintaining grammatical and semantic integrity. Evaluation emphasized whether outputs adhered to the expected matrix–embedded language framework.
- **Comparative Baseline:** To ensure any observed performance issues were not due to implementation errors, select prompts were also tested on Mistral NeMo, providing a cross-model reference point.

These methodologies allowed us to balance replication and exploration: validating prior PANGEA-7B results while also probing its underexplored limitations in handling real-world hybrid language scenarios such as Spanglish.

4 Results

The results are organized into three phases that reflect the progression of our investigation. First, we replicated baseline results reported in the PANGEA-7B paper using the Global-MMLU dataset to validate the reliability of our experimental setup. Next, we examined prompting strategies, including CoT prompting, to assess whether performance on MMLU tasks could be improved. Finally, we evaluated the model's handling of CSW text using a Spanglish dataset, focusing on its ability to generate outputs consistent with linguistic principles of matrix and embedded languages. These stages provide a comprehensive view of PANGEA-7B's strengths and limitations throughout standard multilingual reasoning tasks and real-world hybrid language scenarios.

4.0.1 Replication of PANGEA-7B Results

To establish a baseline, we first replicated results reported in PANGEA-7B's paper. Using the Global-MMLU dataset, we observed accuracy scores averaging between 40–50% on a sample test set of 10 English and Spanish items. Scaling the evaluation to a test set of 100 questions, the model achieved approximately 50–60% accuracy in both languages, which aligns closely with the performance ranges reported in the original PANGEA-7B study, confirming the validity of our experimental setup.

4.0.2 Prompting Strategies

We further investigated prompt design strategies, including CoT prompting, to evaluate whether performance on MMLU tasks could be improved. Contrary to expectations, this approach reduced accuracy, producing scores between 44–54% for both English and Spanish. The model also occasionally generated invalid outputs (e.g., labels outside the A–D range), suggesting a high sensitivity to prompt formulation. Further systematic analysis would be required to confirm and characterize this behavior.

4.0.3 Code-Switching Evaluation

For CSW evaluation, we used Drew Curran's Spanglish dataset, which is structured accordingly to the linguistic principles of matrix and embedded languages. In this framework, English functioned as the matrix language, while Spanish elements constituted the embedded components. When translating English into Spanglish, the model frequently failed to produce outputs consistent with the dataset references. These preliminary findings indicate that prompt design and evaluation configuration substantially influence the model's capacity to generate coherent Spanglish text; however, additional experimentation is required to validate these results.

5 Discussion

Our results highlight both the potential and limitations of PANGEA-7B in multilingual and CSW contexts. On the one hand, the model demonstrated consistent performance in MMLU tasks, achieving accuracy levels comparable to those reported in the original paper. On the other hand, the model underperformed in handling CSW Spanglish text.

These results align with broader findings in the literature that highlight the difficulty of evaluating CSW tasks. The decline in accuracy under CoT prompting underscores the need to tailor evaluation prompts to the architecture of the model.

Additionally, the experiment revealed technical limitations. Coding errors and inconsistencies in evaluation prompts initially interfered with the results, highlighting the importance of robust evaluation pipelines. Once corrected, performance stabilized, but challenges in Spanglish comprehension persisted. These observations suggest that prompt formulation and evaluation setup may play a significant role in the model's handling of CSW text, although further investigation is required to determine the precise causes of these shortcomings.

6 Conclusion

This study investigated the capabilities of PANGEA-7B in multilingual and CSW scenarios, focusing primarily on English, Spanish, and Spanglish. Our replication of reported MMLU results confirmed the reliability of our evaluation pipeline, while our Spanglish experiments revealed persistent challenges in the model handling and generating CSW text.

6.1 Applications

The findings from this study demonstrate both the promise and the current limitations of PANGEA-7B in multilingual and CSW scenarios. By successfully replicating accuracy ranges reported in the original PANGEA-7B paper, this work validates the model's utility for multilingual reasoning tasks such as MMLU and highlights the potential of PANGEA-7B and similar MLLMs for applications in education, cross-lingual information retrieval, and bilingual QA systems where accuracy across English and Spanish is critical.

However, the results from the Spanglish evaluation underscore the gap between monolingual or bilingual performance and real-world hybrid language use. Applications such as automated translation tools, chatbots serving bilingual communities, and digital assistants in English-Spanish contexts would require more robust handling of CSW text than PANGEA-7B currently provides. These findings therefore suggest that future MLLMs intended for deployment in linguistically diverse communities must incorporate CSW-focused training and evaluation to meet real-world demands.

6.2 Concerns and Next Steps

Several concerns emerged during this study. PANGEA-7B's difficulty with Spanglish tasks appeared to stem from its training, and further work would be needed to properly evaluate its performance on CSW tasks. Additionally, prompting strategies had a notable effect on performance: CoT prompting lowered accuracy and sometimes produced invalid outputs. These findings highlight the importance of carefully designing evaluation protocols and prompts to match the architecture of pattern-based models such as PANGEA-7B.

Future work should focus on three areas: (1) refining prompt design strategies tailored to pattern-based models such as PANGEA-7B, (2) expanding evaluation with additional CSW datasets, and (3) exploring the impact of synthetic data generation or fine-tuning on Spanglish-specific corpora. Expanding the scope of evaluation to include other open-source models would also help identify whether the observed limitations are model-specific or reflect broader structural challenges in current MLLMs.

Ultimately, improving MLLM performance in Spanglish and other CSW languages requires innovations at both the data and evaluation levels. By addressing these gaps in data availability and by developing evaluation metrics sensitive to hybrid language dynamics, future work can help build models that more accurately capture the complexity of multilingual communication.

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